

NeuroScan AI

Detecting Approaches to Brain Tumor Classification by Deep Learning and MRI Imaging Paradigm

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This project report is submitted to the Department of Computer Science at Umm Al-Qura University in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Engineering/Computer Science/Information Systems.

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ABSTRACT

Accurate diagnosis of brain tumors from MRI scans is challenging due to the wide variability in tumor shape, size, and location, as well as the limitations of manual interpretation. This project presents NeuroScan AI, an automated deep learning system designed to classify brain MRI images into four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor. The system employs Convolutional Neural Networks (CNNs) with Transfer Learning using architectures such as VGG16 and ResNet50, supported by preprocessing techniques including normalization, resizing, and data augmentation to address data imbalance and enhance model performance. Evaluation metrics—Accuracy, Precision, Recall, and F1-Score—were used to validate the model’s effectiveness. Overall, NeuroScan AI aims to provide a fast, reliable, and explainable tool that assists clinicians in early brain tumor detection and supports more efficient decision-making.

Keywords: Machine Learning, Brain Tumor Detection, Medical Imaging, Deep Learning, MRI, Classification.

Full Term	Abbreviation
Artificial Intelligence	AI
Machine Learning	ML
Deep Learning	DL
Convolutional Neural Network	CNN
Magnetic Resonance Imaging	MRI
Transfer Learning	TL
Support Vector Machine	SVM
Random Forest	RF
You Only Look Once	YOLO
Gradient-weighted Class Activation Mapping	Grad-CAM
Explainable Artificial Intelligence	XAI
Accuracy	Acc
Precision	Prec
Recall	Rec
F1-Score	F1

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Chapter 1 | INTRODUCTION

1.1 Introduction

Diagnosing brain tumors represents one of the most challenging but urgent procedures across medicine, as rapid and reliable identification is critical in determining treatment success and ultimately patient outcome [6]. The greatest challenge lies in leaping over the courage, not named machine learning, to analyze Magnetic Resonance Imaging (MRI) scans, the conventional imaging method, and the well-established artefact of tumor morphological variation in size, shape, and location across patients [7]. The interpretation of the images from MRI scans is a laborious process requiring expert intervention. Moreover, interpreting imaging is prone to being time-consuming, corroborating a need for machine-based help [6, 8]. To this end, Artificial Intelligence (AI) and, more specifically, Deep Learning (DL) has emerged as transformative solutions for the field of medical image -analysis. Convolutional Neural Networks (CNNs) have demonstrated to be the most successful model, as they are uniquely capable of extracting complex image features making them well-suited for the classification of common brain tumors such as Glioma, Meningioma, and Pituitary Tumours. Despite success in deep learning models, a roadblock to developing deep learning models is tumorous heterogeneity, in addition to Class Imbalance in clinical datasets, which can considerably impact model performance, accuracy, and usefulness [7-8]. The objective of this project, NeuroScan AI, is to create an improved deep learning system that can classify brain MRI images into four classes: Glioma, Meningioma, Pituitary Tumour, and No Tumour. The system will help overcome the challenges associated with variability in tumours and imprecision in data due to imbalanced classes through advanced CNN architectures, transfer learning (e.g. VGG16 or ResNet50), and possibly more advanced data preprocessing. The main aim is to provide clinicians with a faster and more efficient preliminary diagnostic tool, thereby facilitating more efficient decision making and improved overall patient care [6].

1.2 Aim

The main goal of the NeuroScan AI project is to develop an automated deep learning system capable of accurately classifying brain MRI images into four categories: Glioma, Meningioma, Pituitary Tumour, and No Tumour, while addressing challenges related to tumor heterogeneity, category imbalance, and model interpretability. The goal is to provide clinicians with a fast, reliable, and explainable preliminary diagnostic tool that can support and accelerate medical decision-making in real-world clinical settings.

1.3 Project Statement

Diagnosing brain tumors with MRI is complicated by the difference in the shape, size, and location of the tumor between patients. MRI data is often unbalanced, making models biased and reducing their ability to generalize. Manual examination is also time-consuming and depends on the experience of doctors.

Although deep learning has been successful, most current models lack clear explanation, reducing doctors' confidence in automated results.

1.4 Project Scope

This project aims to build an intelligent system that can classify brain MRI images into four groups: Glioma, Meningioma, Pituitary Tumour, and No Tumour. The work includes many steps such as collecting medical images from open and trusted sources, cleaning and preparing the data, and improving it using normalization, resizing, and data augmentation to make the model more accurate and avoid overfitting. After that, deep learning models will be built and trained using Convolutional Neural Networks (CNNs). The project will also use transfer learning with models like VGG16 and ResNet50 to improve accuracy and reduce training time. The trained models will be tested and compared with other classifiers such as Support Vector Machine (SVM) and Random Forest to understand which model performs better. Model performance will be measured using accuracy, precision, recall, F1-score, and confusion matrix. At the end, the project will produce a trained and reliable model that helps doctors to make faster and more confident decisions in early diagnosis of brain tumours.

1.5 Project Objectives

The project has several scientific and practical goals: To develop a deep learning model that can classify brain MRI images into four main classes (Glioma, Meningioma, Pituitary, and No Tumor). To improve the data quality by using preprocessing and data augmentation to make the dataset more balanced and stronger. Comparing different models like CNN, VGG16, and ResNet50 to find the best one for brain tumor classification. To evaluate the models using performance metrics such as accuracy, precision, recall, and F1-score to make sure the results are reliable. To create an intelligent tool for doctors that gives a fast and helpful early diagnosis, saving time and supporting better healthcare. To document all project steps (data collection, training, evaluation, and results) clearly in a final report that shows the full process and findings.

1.6 Project limitation

Because the project falls within the healthcare sector, there are privacy and patient information protection restrictions, leading to a lack of information, in addition to data quality limitations such as poor image accuracy or misclassification. Other limitations include technical constraints such as difficulty handling large data volumes or complex models, and human constraints such as limited expertise in medical image analysis.

1.7 Project Organization

The remaining part of the research is organized as follows: Section two provides a comprehensive background on the research and recent studies on the detection of brain tumour types (Glioma, Meningioma, Pituitary, and No Tumor) using various techniques, highlighting their differences. The research then moves to section three, which discusses and explains the data used for clarification. Section four presents the proposed methods such as CNN, VGG16, and ResNet50, explaining the steps of image preprocessing and feature selection. Section five showcases the achieved results, analyzing the model's performance and comparing it with previously used methods. Section six concludes the research, summarizing the main findings, suggesting future directions, and highlighting the contributions of this study. The research will end with a presentation of all the sources used.

Chapter 2 |BACKGROUND

Brain tumours are among the most serious neurological diseases due to their high mortality rates and lasting impact on brain functions, particularly in aggressive cases such as glioblastoma, with statistics indicating that around two-thirds of adult patients with this type lose their lives within two years of diagnosis. Glioma tumours constitute the vast majority of primary malignant brain tumours, while meningioma and pituitary tumours tend to be more stable and less malignant, though they remain dangerous due to their sensitive locations and hormonal or neurological effects[2]. This highlights the importance of relying on artificial intelligence for faster and more accurate diagnosis, CNN-based models can accurately extract fine patterns from MRI images, outperforming traditional manual diagnostic methods.. Studies show that Transfer Learning techniques such as pre-trained ResNet-18 increase prediction accuracy to 98–99% in classifying images into the four categories (glioma, meningioma, pituitary, no tumor). Explanation algorithms (such as Grad-CAM) also help identify the areas the model relied on in making decisions, enhancing its reliability and assisting doctors in interpreting the results[1]. To achieve optimal performance, research emphasizes the importance of pre-processing steps such as noise removal, image dimension standardization, and colour value normalization, in addition to applying segmentation techniques to separate the tumour mass from healthy tissue for extracting precise morphological and histological features[2].

2.1 Artificial Intelligence & Deep Learning

Artificial Intelligence (AI) and Deep Learning (DL) have evolved significantly in recent years, enabling computer systems to automatically learn visual features from MRI images without the need for manual feature extraction .This project (NeuroScan AI) includes the use of advanced deep learning architectures (VGG16, ResNet50) to categorize brain MRI images into four classes: Glioma, Meningioma, Pituitary No Tumour. To effectively solve the problem of limited and imbalanced medical data, this project employs transfer learning procedures for increasing the accuracy of the model while improving training efficiency. In addition to transfer learning, data augmentation and normalization techniques are applied to improve model performance and limit overfitting. Explainable AI (XAI) methods (especially Grad-CAM Gradient-weighted Class Activation Mapping, which highlights the most significant portions of MRI images for classification) are integrated into the project to help improve the transparency of the model and, more importantly, to provide medical experts with greater understanding and trust in the models' classification results. **Our ultimate goal with our project is to provide an accurate and effective diagnostic tool that helps radiologists in the early detection of brain tumours and decision-making.**

2.2 Machine Learning (ML)

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that focuses on teaching machines how to learn from data without being explicitly programmed. The types of machine learning are divided into three main categories. The first type is Supervised Learning, which depends on labeled data. In this process, both the input and output are provided, and the model learns the pattern or relationship between them. Examples of this type include classification and regression. The second type is Unsupervised Learning, which works on unlabeled data. It learns patterns in the data without human intervention and is widely used for feature extraction, with one of the most common techniques being clustering. The third type is Semi-Supervised Learning, which is a hybrid between supervised and unsupervised learning. It works on both labeled and unlabeled data, since in real life, it's rare to find fully labeled datasets. The goal of this approach is to improve results by combining both methods. In short, the way machine learning works can be summarized in two main stages: first, the model is trained using historical data, and second, it is tested to check its accuracy and ability to make correct predictions. Deep Learning (DL) is considered a part of Machine Learning (ML). It relies on the use of Artificial Neural Networks (ANNs), which are systems made up of multiple hidden layers that help analyze data more deeply and accurately. DL algorithms are among the most widely used and are commonly applied in pattern recognition systems, as they can identify complex relationships within data. Deep learning techniques have proven successful in handling challenging types of data, such as high-dimensional, noisy, and unstructured datasets like images and text. Moreover, deep learning is widely used in many fields, especially in healthcare, where it plays an important role in analyzing medical images and detecting tumors.

2.3 The role of the of Saudi Arabia in implementing artificial intelligence in the healthcare sector

The growing role of artificial intelligence (AI) in the development of Hospital Management Information Systems (HMIS), its role in improving the quality of health services, increasing resource management efficiency, and supporting evidence-based clinical decisions. Moreover, integrating AI into the healthcare sector has become a pivotal step to improve patient outcomes and reduce medical errors through applications that include disease prediction, medical image analysis, automated diagnosis, and natural language processing (NLP), in addition to Clinical Decision Support Systems (CDSS) that help doctors make accurate and immediate decisions. Saudi Arabia has also emphasized the importance of adhering to ethical and religious frameworks in AI applications within the country, as national policies are based on the principles of Islamic law, which uphold privacy and human dignity. It is important that AI should complement human medical decisions, not replace them. While AI facilitates quick and data-driven

decision-making, institutions must ensure that these decisions are accurate, transparent, and unbiased to maintain public trust and sustain the digital transformation in the healthcare sector.[5]

2.4 Types of brain tumours

This project focuses on classifying three categories of brain tumours in addition to the no-tumour category, as these categories represent clear differences in nature and location between malignant and benign tumours.

2.4.1 No Tumor (Normal Brain)

The human brain is the main control center of the body. It has billions of nerve cells that send messages through a fast network of connections. These signals help us move, feel, think, and remember. The largest part is the cerebrum, which has two halves — left and right — connected by the corpus callosum. The outer layer, called gray matter, helps with thinking and understanding, while white matter connects different parts of the brain. Each half has four lobes:

Brain Lobe	Frontal lobes	Parietal lobes	Occipital lobes	Temporal lobes
Main Functions	for planning, movement, and problem solving	for touch and sensing	for vision	for hearing, memory, and emotion

Below the cerebrum, the cerebellum controls balance and coordination, while the brainstem manages automatic actions like breathing and heartbeat. Inside the brain, the limbic system helps control emotions and memory. It includes the thalamus, hypothalamus, and hippocampus. All parts of the brain work together with the nervous system to control everything we do — from simple reflexes to complex thoughts. [23] Figure 1 shows a basic MRI scan of a brain without any illness. This normal scan serves as the perfect example of what the classification system should see as healthy. The image is clear, and the different parts of the brain (like the gray and white areas) are separated correctly. Most importantly, there is no sign of any abnormal mass or any area where the tissue looks swollen or out of place. This clear, consistent appearance is what we look for when we say a patient has "No Tumor". [12]

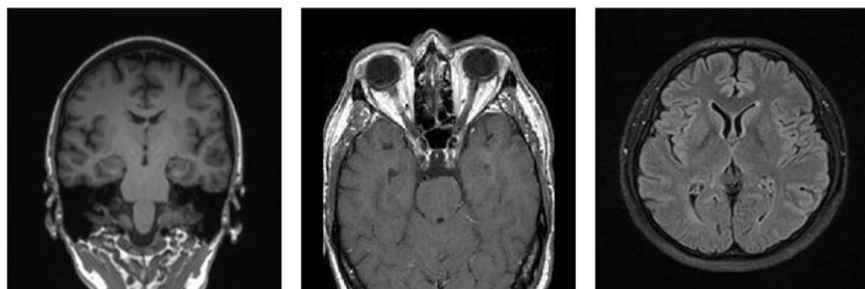


Figure (1): Standard Magnetic Resonance Imaging (MRI) View of a Healthy Brain. [12]

2.4.2 Glioma

A glioma is a type of tumor that starts in the brain's glial cells, which normally support and protect the nerve cells. It can grow and spread inside the brain. Gliomas commonly happen in areas where brain cells are very active, and they may hijack the brain's normal communication system to help the tumor grow. [24] Because gliomas merge into normal brain tissue, they affect many functions: they may cause seizures, trouble with thinking, or problems with movement or speech depending on the part of the brain they invade. Treatment and outcome vary with tumor type and how early it is found. Some gliomas grow slowly, while others are fast and aggressive. The network of tumor-cells and their interaction with nerve cells are important for how the disease behaves. [25] The image in Figure 2 is an MRI T2 scan used to find a Glioma. This tumor starts from the support cells in the brain. On this type of scan, the glioma shows up as a remarkably bright white spot. This bright color means there is a lot of fluid in the tumor and in the area around it (swelling). A major feature of this tumor, which is clear here, is that its edges are not sharp or clean. This happens because the tumor cells are growing and spreading into the good brain tissue nearby. The classification model learns to recognize this diffuse, spreading pattern. [12]

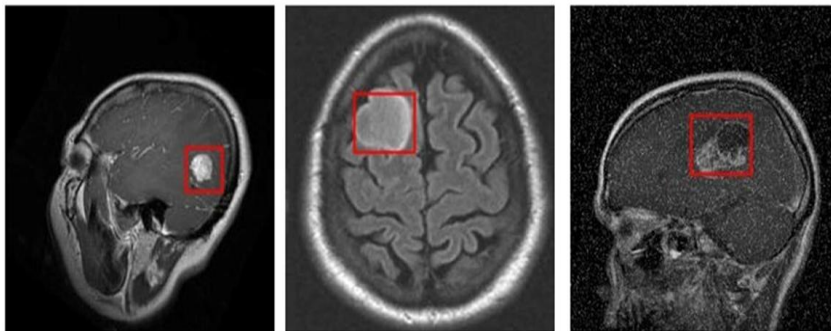


Figure (2): MRI Scan Showing a Glioma. [12]

2.4.3 Meningioma

Meningiomas are the most common primary brain tumors that arises from the meninges (the membranes that cover the brain and spinal cord).[9-10] Most are benign (non-cancerous) and slow-growing; however, some can be atypical or malignant (cancerous). The accompanying MRI images in Figure 3 depict a meningioma in three planes: axial (left), sagittal (middle), coronal (right). Red boxes have been placed in each image to indicate the location of the tumor. The meningioma appears as a well-defined extra-axial mass that is attached to the what's called the dura mater and has higher signal intensity and homogeneous enhancement on post-contrast (gadolinium) images. One common feature of meningiomas is the "dural tail sign" (the enhancement of the adjacent dura that tapers), which supports the diagnosis of a meningioma. Mild edema is often seen around the lesion reflecting mild compression on the adjacent brain. Overall, this

is the classic appearance of a meningioma: a dural-based, well-defined lesion that shows uniform enhancement with contrast medium, which assists in diagnosis and treatment. Differential diagnosis should consider other extra-axial masses such as metastases or schwannomas, but the dural attachment and homogeneous enhancement strongly favor a meningioma. Clinically, these tumors may cause symptoms such as headaches, seizures, or focal neurological deficits depending on their size and location. MRI plays a crucial role not only in diagnosis but also in surgical planning, as it helps to delineate tumor boundaries, evaluate involvement of adjacent structures, and assess potential recurrence after treatment.

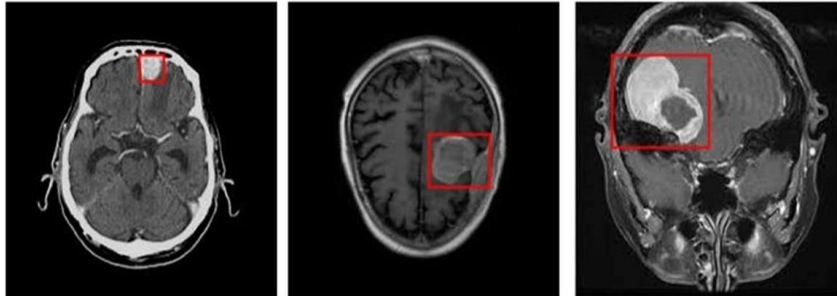


Figure (3): Enhanced MRI Scan of a Meningioma. [12]

2.4.4 Pituitary

Pituitary tumors are among the most common brain tumors, accounting for about 10–15% of all brain tumors. They come from the anterior pituitary gland, which is important for many hormone functions in the body. Most pituitary tumors are benign and called pituitary adenomas (PAs). Only 0.1 to 0.2% are malignant and classified as pituitary carcinomas (PCs). Most pituitary adenomas grow slowly and are often found by chance during magnetic resonance imaging (MRI) done for other reasons; these are known as pituitary incidentalomas. These tumors usually stay the same size and do not need treatment unless they grow or cause symptoms related to hormonal imbalance or pressure on nearby structures. However, about 10% of pituitary adenomas show aggressive or stubborn behavior. This helps in understanding tumor subtypes and behaviors more clearly. Even though most pituitary tumors are benign, early diagnosis and classification are still crucial to avoid complications and improve treatment results. [26] Figure 4 presents an important view of a Pituitary Tumor using an MRI scan. The picture clearly highlights an abnormal growth. This growth is a well-defined mass, meaning it has clean, sharp edges. This is different from tumors that spread out into the brain tissue. The most important feature is the mass's specific location in the Sella region, which is where the pituitary gland sits. The tumor appears compact and smooth. Its size, as seen in this scan, is large enough to show that it is pressing on the structures around it. This visual pattern—a regular shape found in a central, fixed spot—is a key finding when studying this type of tumor.

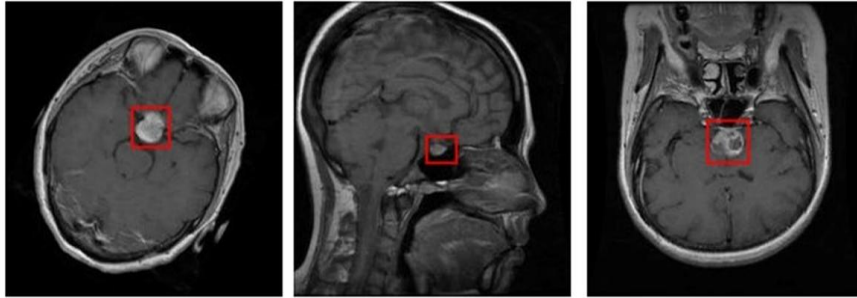


Figure (4): MRI Scan of a Pituitary Adenoma (Tumor). [12]

Chapter 3 | Literature review

Based on these references [3], both YOLO-8 and YOLO-11, the latest version of the YOLO family, were used. This version offers the newest architecture and greater efficiency. YOLOv11 is designed to enhance small object detection and improve accuracy while maintaining the real-time inference speed that YOLO models are known for. Finally, the SiLU activation function is used instead of the Sigmoid function. The adjustment processes for YOLO v8 and YOLO v11 models involved several stages similar to those applied to previous models. The specific features and characteristics of their training processes are detailed in the table below.

Epochs	20
Image Size	224x224
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Augmentation	Enabled
Device	GPU (device0)

Table (1): YOLOV8 & YOLOV11 Training characteristics.[3]

YOLO remains one of the most robust and reliable pre-trained infrastructures for brain tumor detection, demonstrating its superior performance in this field. Specifically, YOLOv8 and YOLOv11 stand out as the most effective pre-trained models for MRI classification tasks, with accuracies of 99.31% and 99.50%, respectively. These models have shown exceptional accuracy and robustness, setting a benchmark for medical imaging applications.

In 2022, the research paper [2] presented a proposed reliable approach for early tumor classification with high accuracy compared to manual methods. The steps followed here are: **(1) MRI image collection**; they used images of human brains at horizontal levels, **(2) preprocessing**; which aims to obtain cleaner and clearer images to increase accuracy, **(3) feature extraction**; extracting characteristics that distinguish normal tissue from tumorous tissue, which are later used for classification, **(4) image segmentation**; where several techniques were used for comparison such as *< Threshold_based Segmentation – Region_based Segmentation – Edge_based Segmentation – Fuzzy_based Segmentation – NN_based Segmentation >*, **(5) post-processing**; applying shape and spatial constraints to improve the final result, **(6) classification**; which relies on the features extracted from the previous steps.

In [1], lightweight deep learning models were developed to classify brain tumors in MRI images into four main categories. The researchers used a dataset consisting of 7,023 MRI images, of which 5,712 were for training and 1,311 for testing, collected from various public sources to ensure the model's comprehensiveness and generalization capability. The study began with a preprocessing stage for the images, where all images were resized to 224×224 pixels while maintaining aspect ratio, followed by a series of augmentation and enhancement operations. After that, five convolutional neural network (CNN) models were selected and trained. The results showed that the pretrained ResNet-18 model achieved the best performance among all models, recording a classification accuracy of 98.86%. To enhance the transparency of the decisions made by the models, the researchers used Explainable AI techniques such as Grad-CAM and Occlusion Maps to highlight the regions the model relied on for decision-making. In conclusion, the study demonstrated that using lightweight models based on transfer learning, such as ResNet-18, can achieve performance comparable to or exceeding that of large models, with high computational efficiency making them suitable for deployment in resource-limited medical environments.

Based on the literature [8], the paper “Hybrid Transfer Learning and Self-Attention Framework for Robust MRI-based Brain Tumor Classification” proposed a hybrid deep learning framework called DenseTransformer to improve classification and detection of brain tumors from MRI images, which combines DenseNet201 for spatial feature extraction with Transformer-based Multi-Head Self-Attention (MHSA) and Squeeze-and-Excitation Attention (SEA) modules, both enhancing classification performance and feature representation. The model employed a dataset entitled Br35H MRI dataset consisting of 3,000 technical images, and, in its implementation, performed preprocessing steps including normalization, resizing to 224×224 , and data augmentation to improve model accuracy and balance. The model reached a very respectable accuracy of 99.41%, achieving higher accuracy compared to well-known models such as VGG19, ResNet50, MobileNetV2, and Inception-V3. In addition, quantitative analyses performed using a Z test and a McNemar’s test validated the enhanced performance of the model.

Explanatory AI strategies like Grad-CAM and LIME were also personalized to visualize areas that might be important to the model predictions, in the interest of model interpretability and clinical transparency. In summary, the combination of using transfer learning and the models of attention yields accuracy and interpretation for MRI-based classification of brain tumors, demonstrating the potential for future clinical applications.

Based on the review article “A Review of Deep Learning for Brain Tumor Analysis in MRI” [6],

We conclude that deep learning has become a key tool in brain tumor diagnosis and analysis, focusing mainly on segmentation, quantification, and classification tasks.

(1) Tumor Segmentation and Quantification; Segmentation plays an essential role in diagnosis and treatment planning, and most recent methods depend on CNN architectures like U-Net and nnU-Net. These models automate much of the training process and have become a standard for medical image segmentation. Model performance is usually evaluated using the Dice Score, and the BraTS challenge remains the main benchmark in this field. Quantitative DL models also provide more accurate and reproducible tumor volume measurements compared to manual work and are used in planning radiotherapy. Managing prediction uncertainty is another important step to increase clinical reliability.

(2) Classification and Clinical Prediction; Deep learning classification models, especially from the ResNet family, are used to identify tumor types and predict genetic features such as IDH and MGMT status. They can also distinguish between tumor recurrence and radiation necrosis. Moreover, some models predict patient survival time, supporting more personalized treatment planning.

(3) Future Trends; The new generation of DL models focuses on improving generalization and clinical impact. Foundation models trained on large datasets, along with the integration of imaging, clinical, and genetic data, are expected to push accuracy even further. In addition, quantitative MRI (qMRI) aims to standardize image quality across scanners and provide deeper biological insights into tumor behavior.

Based on Reference [7], The study by Khan et al. (2025) talked about using transfer learning to improve how brain tumors are classified in MRI images. They used three well-known deep learning models: AlexNet, MobileNetV2, and GoogleNet, to see which one gives the best results when identifying different types of brain tumors. The dataset they used was from Kaggle Brain MRI Dataset, which includes 4,517 MRI images divided into four types: glioma, meningioma, pituitary tumor, and no tumor. Before training, they cleaned and prepared the data by adjusting brightness, flipping, rotating, and resizing the images. They also split the dataset into 80% for training and 20% for testing. After testing all three models, GoogleNet gave the best accuracy at 99.2%, followed by MobileNetV2 with 98.7%, and AlexNet with 96.5%. GoogleNet was better in detecting small and complex details in the brain images. Its results were higher than other studies that used models like VGGNet or ResNet34. The researchers mentioned that MobileNetV2 is a good option because it's fast and doesn't need strong hardware, which makes it suitable for hospitals that don't have high-end devices. They also suggested using Explainable AI tools like Grad-CAM to show which parts of the MRI image the model focused on, making the prediction easier to

understand. In general, the study showed that transfer learning can make brain tumor classification more accurate and reliable. The models, especially GoogleNet, can help doctors detect tumors faster and with more confidence. The authors also said that future research should use bigger and more varied datasets to make the models work better in real hospitals.

In this paper [11], we propose an advanced model for brain tumor classification with the use of Magnetic Resonance Imagery (MRI), relying on the improved version of the EfficientNet-B0 network. The problems of brain tumor diagnosis are associated with the complex structures in pathology diversity of medical image data. To address this bottleneck, the authors incorporated the Multi-Scale Channel Attention Mechanism (MS-CAM) into MBConv to help the model better aggregate multi-scale feature information. Experiments The proposed model was trained and tested on a dataset consisting of 7023 MRI images, and a classification accuracy of 99.31% was observed over the validation set. Compared to typical models, like EfficientNet-B0, GhostNet, and MobileNetV3, the improved model of EfficientNet-B0 has outstanding performance with respect to the accuracy and robustness of detection of brain tumors. Such results show its promising use as an efficient AI-assisted medical diagnosis and healthcare tool.

In this paper [12] proposed a new way to classify the brain tumors using an ensemble of deep learning model combined with ResNet50 and EfficientNet-B7. This model uses high resolution MRI images to distinguish four types of tumour: no tumour, glioma, meningioma and pituitary tumour. A dataset of more than 22,000 MRI images from the “Crystal Clear: Brain Tumors MRI Data set” was employed for training and validation. The combined method consolidates the results obtained by both the models using weighted averaging in which weights are decided according to their training performance and thus to improve the classification accuracy. From the experiments, it was revealed that the stand-alone ResNet50 model got a validation accuracy of 97.65% and EfficientNet-B7 achieved an accuracy of 98.20%. Interestingly enough, the ensemble model performed better than two individual models with validation accuracies of 99.68% and 99.53%, after training. This trial ensemblment model demonstrated an excellent accuracy and generalization in distinguishing these similar tumor types (e.g. glioma-meningioma) with low misclassification rates. It demonstrates the effectiveness of ensemble learning in improving classification performance and generalization, which facilitates accurate and reliable diagnostic tool for computer-aided diagnosis on brain tumor detection in medical imaging.

In this paper, [13] Res-BRNet, a new deep learning model to tackle the issue of brain tumour classification with Magnetic Resonance Imaging (MRI). This model successfully combines spatial (regional) and residual convolutional blocks, which preserve rich tumor features for better classification. The authors used data obtained from Kaggle, Br35H, and Figshare, consisting of MRI images that have four different classes: healthy brain, glioma, meningioma, and pituitary tumor. To increase the generalisation of the model as well as address data imbalance, the team applied a number of augmentation techniques such as random rotation, scaling, and flipping. The Res-BRNet model obtained a classification accuracy rate of 98.22%, which was competitive with the previous methods. at a sensitivity of 0.9811 and F1-score of 0.9641, surpassing several state-of-the-art architectures including VGG-16, ResNet-18 and DenseNet-201. Assessment based on confusion matrices and ROC–AUC curves showed that the model has high discrimination power with minimal misclassification error. The results show that Res-BRNet is a robust and accurate AI-based imaging diagnostic tool for brain tumor detection, which can substantially improve diagnostic reliability and decrease clinicians' burden. The paper underscores the value of the model to improve deep learning in medical imaging and diagnostic accuracy for brain tumors.

In this paper, [14] compared several approaches for brain tumor classification from MRI images in a full comparative study, including machine learning and deep learning techniques. The models used in the experiment were Convolutional Neural Networks (CNN), Support Vector Machine (SVM), Random Forest Classifier (RFC), and transfer learning models [34, 35] like VGG16, MobileNetV2, InceptionV3, and Xception. Also, an adaptive fusion of MobileNetV2 with SVM was created for efficient learning and classification. The dataset comprised 7,023 MRI images in four classes (meningioma, glioma, pituitary tumor, no tumor), with an 80/20 train-test split. All images were resized to 180×180 pixels and converted to grayscale, and they were augmented by random rotation, flip, zoom, and Gaussian noise for generalization. We found that the combination of MobileNetV2 and SVM provided the highest classification accuracy, 98.2%, followed by VGG16 (97.9%), CNN (97.6%), SVM (95%), and the RFC model (94%). CNN proved highly robust, in particular when tested on low-quality MRI images. In addition, the models performed better than radiologists, whose average accuracy was 87%, especially for distinguishing similar tumor types. This led to the conclusion that a MobileNetV2/SVM hybrid model is striking an optimal trade-off between accuracy and computational cost, making it applicable in practice for medical diagnostics systems and deployment close to the field, such as low-magnetic field MRI scenarios.

In this paper [15] introduced an end-to-end deep learning approach for an automated classification and segmentation of brain tumors using Multiscale Convolutional Neural Network (Multiscale CNN). The MRI images are fed into three parallel paths of the model, which is special in that features at large scale, medium scale and small scale are extracted separately so as to mimic (HVS) feature perception. We used 233 samples and 3064 MRI slices in our study, which were divided into three tumor types including meningioma, glioma and pituitary tumor. The information was split into 80% training data and 20% test sample, while the robustness is obtained through a 5-fold cross validation. Unlike many of the existing methods, the model did not require pre-processing to remove non-brain compartments such as the skull or spinal cord. During training, we used the elastic transformation for data augmentation in order to enhance model generalization and reduce overfitting. The performance of the developed M-CNN (97.3% in classification accuracy, 0.828 in Dice coefficient, 0.940 in sensitivity and 0.967 PPTAS) improved over conventional techniques including SVM and baseline CNN method. Moreover, it performed better than the trained radiologists that had an accuracy of 91.2%. In general, our study showed that the Multiscale CNN is a promising and efficient solution for automatic brain tumor classification and segmentation. providing a reliable AI-based solution to enhance diagnostic accuracy and support medical professionals in clinical decision-making.

[16] This paper proposes the MAG-Net model, which is based on the concept of multi-task learning, allowing the model to perform two tasks simultaneously: Classification to determine the tumor type and Segmentation to extract the tumor's shape and boundaries. The main idea is to integrate Attention Mechanisms within a simplified Encoder-Decoder network so that the model can focus on important areas of the brain, which helps make the model more effective with fewer data. An important feature here is the attempt to unify the tasks of segmentation and classification in a single model. The results were strong, especially in classification, with high accuracy and a reduction in the number of parameters compared to similar models.

In this paper [17] To help doctors determine the appropriate treatment before the tumor grows, they used a Deep Neural Network model for early tumor detection, which implements two main stages: Detection to identify the presence of a tumor & Segmentation to precisely locate the tumor area within the MRI image. The paper aims to address common challenges such as tumor size, location, shape, the diversity of medical devices, and variations in their quality. Therefore, they used preprocessing techniques such as noise removal and normalization to improve the quality of the model's input. The model produced very good results in tumor detection and achieved high Dice Score values during tumor segmentation, indicating high accuracy

in defining tumor boundaries. The paper focused more on practical applications in early medical interventions rather than just the technical aspect.

Based on Reference [18], the study "**Lightweight CNN for accurate brain tumor detection from MRI with limited training data**" proposed a highly efficient and simple Convolutional Neural Network (CNN) architecture. This architecture was specifically designed for the binary classification of MRI images—distinguishing between cancerous (tumor) and non-cancerous categories—even with a limited amount of training data. The proposed CNN consists of five optimized layers, which helps reduce computational complexity while maintaining strong performance. The experimental results confirmed the efficacy of this lightweight approach, achieving a significant classification accuracy of **96.11%**. This demonstrates the feasibility of developing highly accurate detection systems that are practical for deployment in environments where data or computational resources may be constrained.

Based on these references [19], in 2025, the research paper "**Applying Deep Learning Techniques in Accurate Brain Tumor Detection and Classification**" demonstrated the significant enhancement of diagnostic efficiency and accuracy through deep learning. The authors classified and identified brain tumors using four deep learning models: a standard Convolutional Neural Network (CNN), VGGNet19, ResNet101V2, and EfficientNetV2B2. The classification included four distinct categories: no tumor, glioma, meningioma, and pituitary. The study showed that the CNN and VGGNet19 models displayed exceptional performance, with **VGGNet19 achieving the highest score of 99.81%**. This superior performance, coupled with high stability, makes these models highly suitable for practical clinical applications, ultimately supporting improved patient treatment outcomes.

In 2025, the research paper [20], titled "**Advanced Brain Tumor Classification in MR Images Using Transfer Learning and Pre-Trained Deep CNN Models**", focused on leveraging advanced deep learning and transfer learning techniques to effectively classify brain MRI images. The core aim was to enhance the accuracy and efficiency of medical imaging systems by testing state-of-the-art pre-trained models, specifically Xception, MobileNetV2, and InceptionV3. These models were fine-tuned using a publicly available Brain Tumor MRI dataset, which categorized images into four distinct classes: Pituitary, Meningioma, Glioma, and No Tumor. The experimental results clearly demonstrated that the **Xception model outperformed the others, achieving a weighted accuracy of 98.73%**. This study confirms that

transfer learning significantly conserves computational resources and training time while boosting performance, offering a powerful tool for supporting professionals in cancer diagnosis.

In this paper [21], the researchers introduced a robust, automated system specifically designed for brain tumor classification, leveraging the immense capabilities of **Deep Learning (DL)** on Magnetic Resonance Imaging (MRI) scans. The principal aim of this critical study was to construct an exceptionally reliable model that could accurately perform **multi-class classification**, distinguishing between four distinct tumor types: **glioma, meningioma, pituitary tumors, and normal brain scans**. The team achieved a remarkably high classification accuracy of **99.24%**. This outstanding result was primarily attributed to their meticulous methodology, which focused on three major, interconnected strategies: first, the strategic use of a **large and diverse dataset** to ensure the model's generalization; second, significant **improvements in the underlying network configuration** to enhance learning efficiency; and third, the effective application of a **fine-tuning strategy** to optimally adjust pre-trained weights for the medical task. Ultimately, the authors concluded that this proposed system offers an extraordinarily efficient and trustworthy solution, marking a substantial advancement in medical imaging technology that directly facilitates prompt early diagnosis and allows for more timely medical interventions.

Based on Reference [22], a compelling study was meticulously carried out to thoroughly **evaluate and compare the performance** of various deep learning models dedicated to brain tumor classification, placing a significant emphasis on the advanced **YOLOv8s-cls** model. The YOLOv8s-cls model was strategically employed with the primary objective of directly and accurately detecting brain tumors from MRI scans, aiming to establish an impressively high success rate in the medical image classification domain. The effectiveness and robustness of this specific model were rigorously evaluated using a comprehensive public dataset that encompassed over 3,000 images, where it achieved a noteworthy high accuracy rate of **98.7%** during the extensive experimental trials. Crucially, the authors emphatically highlighted that the sophisticated YOLOv8 model not only demonstrates clear superiority over many traditional detection methods but has also been uniquely optimized for a dual focus on both **classification speed and overall effectiveness**. This distinct capability, particularly its proven potential for **real-time processing**, definitively establishes the YOLOv8s-cls model as an incredibly valuable asset, suitable for rapid clinical deployment to significantly enhance early tumor detection.

NO.	Paper Title	Year	Technique	Dataset	Preprocessing Step	Accuracy (%)
[3]	Detecting Glioma, Meningioma, and Pituitary Tumors, and Normal Brain Tissues based on Yolov11 and Yolov8 Deep Learning Models	2024	YOLOv8 & YOLOv11 (Pre-trained Object Detection Models)	MRI Images	Resize to 224×224, Augmentation Enabled	YOLOv8: 99.31 YOLOv11: 99.50
[2]	An Analysis of Image Segmentation Methods for Brain Tumour Detection on MRI Images	2022	Multi-stage Traditional + NN-based Segmentation	MRI Images (Human Brain, Horizontal Levels)	Preprocessing for noise removal, Feature Extraction, Segmentation (Threshold / Region / Edge / Fuzzy / NN), Post-processing	-
[1]	Lightweight Transfer Learning Models for Multi-Class	2023	ResNet-18 (Transfer Learning, CNN-based)	7,023 MRI Images (Public Sources)	Resize to 224×224, Augmentation, Enhancement	98.86
[8]	Hybrid Transfer Learning and Self-Attention Framework for Robust MRI-based Brain Tumor Classification	2024	DenseNet201 + Transformer-based MHSA + SEA (Hybrid Deep Learning Framework)	Br35H MRI Dataset (3,000 images)	Normalization, Resize to 224×224, Data Augmentation	99.41
[6]	A Review of Deep Learning for Brain Tumor Analysis in MRI	2023	Deep Learning Models (CNNs, U-Net, ResNet Family)	MRI Images (BraTS Challenge)	Segmentation, Quantification, and Classification Tasks	-
[7]	Transfer Learning Models for Brain Tumor Classification in MR Images	2025	AlexNet, MobileNetV2, GoogleNet	Kaggle Brain MRI Dataset (4,517 Images)	Brightness Adjustment, Flipping, Rotation, Resizing (224×224)	GoogleNet: 99.2

[11]	An Improved EfficientNet-B0 Approach for Brain Tumor MRI Image Classification	2025	Improved EfficientNet-B0 with MS-CAM Attention Mechanism	7023 MRI images (Glioma, Meningioma, Pituitary, No Tumor)	Resize to 224×224, Data Augmentation (translation, rotation, horizontal flip)	99.31%
[12]	Ensemble Deep Learning Models for Enhanced Brain Tumor Classification by Leveraging ResNet50 and EfficientNet-B7 on High-Resolution MRI Images	2024	ResNet50, EfficientNet-B7 (Ensemble Deep Learning Model)	Crystal Clean: Brain Tumor MRI Dataset (22,000 high-resolution MRI images)	Resize to 224x224, Data Augmentation, Normalization	ResNet50: 97.65%, EfficientNet-B7: 98.20%, Ensemble: 99.53%
[13]	Brain Tumor MRI Classification Using a Novel Deep Residual and Regional CNN	2024	Res-BRNet (Deep Residual and Regional Convolutional Neural Network)	Kaggle, Br35H, Figshare datasets (four classes: Normal, Glioma, Meningioma, Pituitary)	Resize images, Data Augmentation (rotation, scaling, flipping)	98.22% (Sensitivity: 0.9811, F1-score: 0.9641)
[14]	Brain Tumor Classification in MRI with Deep Learning	2024	CNN, SVM, RFC, Transfer Learning (VGG16, MobileNetV2, InceptionV3, Xception), Hybrid (MobileNet V2 + SVM)	7023 MRI images (Meningioma, Glioma, Pituitary, No Tumor), 80/20 train-test split	Images resized to 180×180 px, grayscale, data augmentation (rotation, flipping, zoom, Gaussian noise)	Hybrid (MobileNet V2/SVM): 98.2%, VGG16: 97.9%, CNN: 97.6%, SVM: 95%, RFC: 94%
[15]	A Deep Learning Approach for Brain Tumor Classification and Segmentation Using a Multiscale Convolutional Neural Network	2021	Multiscale CNN (three parallel convolutional paths for large, medium, and small scales)	3064 MRI slices from 233 patients (M,G,P) collected from hospitals in China	80/20 train-test split, 5-fold cross-validation, elastic transformations for data augmentation	97.3% (Dice: 0.828, Sensitivity: 0.940, PTAS: 0.967)

[16]	MAG-Net: Multi-task attention guided network for brain tumor segmentation and classification	2021	Multi-task deep learning model combining Encoder–Decoder + Attention + Multi-Task Learning	Figshare Brain Tumor MRI Dataset (Meningioma, Glioma, Pituitary)	Normalization	98%
[17]	An early detection and segmentation of Brain Tumor using Deep Neural Network	2023	Deep Neural Network (DNN) + CNN-based segmentation	Public MRI Brain Tumor Dataset	Noise removal (Gaussian filtering)	96%
[18]	Lightweight CNN for accurate brain tumor detection from MRI with limited training data	2025	Lightweight CNN (5-Layer Architecture)	UCI Repository and Kaggle (Binary Classification)	Optimizing CNN architecture	96.11%
[19]	Applying Deep Learning Techniques in Accurate Brain Tumor Detection and Classification	2025	VGGNet19 (Deep CNN)	Four tumor categories (Glioma, Meningioma, Pituitary, No Tumor)	Optimized Feature Extraction	99.81%
[20]	Advanced Brain Tumor Classification in MR Images Using Transfer Learning and Pre-Trained Deep CNN Models	2025	Transfer Learning (Xception)	Publicly available Brain Tumor MRI dataset (Four classes)	Fine-tuning pre-trained models	98.73%
[21]	Brain tumor classification using MRI images and deep learning techniques	2025	Optimized CNN Model	Large and diverse dataset (Four tumor types)	Fine-tuning pre-trained weights, Data Augmentation	99.24%
[22]	Performance Comparison of Deep Learning Models in Brain Tumor Classification	2025	YOLOv8s-cls (Fast Detection)	Public dataset (over 3,000 images)	Optimized for speed and real-time processing	98.7%

[27]	Enhancing brain tumor detection in MRI images through ResNet50 + Grad-CAM	2024	Using ResNet50 + visual interpretation technology (Grad-CAM)	MRI dataset with augmentation enhancement	Data Augmentation, Image Size Standardization, Standard Pixel Intensity Normalization, and Data Cleaning	99.52%
[28]	Brain Tumor Detection and Prediction in MRI Images Utilizing a Fine-Tuned Transfer Learning Model Integrated Within Deep Learning Frameworks	2025	Fine-Tuned Transfer Learning using the Xception model	Brain Tumor Dataset from Kaggle (3762 images)	Applying Data Augmentation techniques (rotation, flipping, scaling) to reduce class imbalance and increase dataset diversity. Images were resized to 150 \times 150.	96.11%

Table (2): Literature review summary.

Chapter 4 | Methodology

4.1 Introduction and Approach Justification

This chapter outlines the detailed **Methodology** that will be followed to implement the **NeuroScan Ai** project. The entire project focuses on developing a highly efficient system for the multi-class classification of brain tumors using advanced Deep Learning (DL) models on high-resolution Magnetic Resonance Imaging (MRI) scans. The methodological framework is specifically derived from the established, high-performing research conducted by Sakr et al. (2024) [12], which successfully demonstrated the use of sophisticated Convolutional Neural Networks (CNNs) for superior diagnostic results. The methodology is structured into distinct, sequential stages to ensure a systematic and reliable system construction.

4.2 Data Acquisition and Management

The success of any DL-based system hinges on the quality and volume of the training data. This phase details how the image data is sourced and prepared for the models.

4.2.1 Dataset Collection and Source

The primary task involves gathering a **large volume of high-resolution MRI images**. This emphasis on image resolution is critical because deep CNN models require minute morphological details to accurately differentiate between various tumor types. The images collected are categorized into the four target classes: **Glioma, Meningioma, Pituitary tumor, and Normal brain scans**. Sourcing data from established public medical repositories ensures a broad and representative collection of pathological cases.

4.2.2 Data Partitioning Protocol

To ensure the models are trained rigorously and tested fairly, the full dataset is strictly partitioned into three distinct, non-overlapping subsets:

1. **Training Set:** This is the largest portion, used exclusively for updating the model's internal parameters (weights and biases).
2. **Validation Set:** This set is used during the training phase to monitor the model's learning progress and to tune hyperparameters, a key step in preventing **overfitting**.
3. **Test Set:** This is reserved as a final, completely unbiased set of data. It is only used once, at the very end of the process, to assess the system's true ability to generalize to new patient cases.

4.3 Image Preprocessing Techniques

Raw MRI images contain variations that can hinder model training, making preprocessing a mandatory step to standardize the input data.

4.3.1 Standardization (Resizing and Normalization)

All high-resolution MRI images must be transformed into a consistent format suitable for the models. This involves two major steps:

- **Uniform Resizing:** Images are scaled to a fixed pixel dimension (e.g., 224x224 or 300x300), which is dictated by the specific input layer requirements of the deep CNNs used.
- **Intensity Normalization:** The raw pixel intensity values are rescaled across all images into a small, standardized numerical range (typically 0 to 1). This ensures that no single image's brightness or contrast unduly affects the model's training convergence.

4.3.2 Advanced Data Augmentation

To significantly increase the robustness of the models and effectively mitigate the risk of **overfitting** to the limited training samples, **Data Augmentation** is performed. This technique involves generating new, synthetic training variations by applying various random yet realistic transformations to the original images. Common augmentations used include:

- Random rotations and minor scaling (zoom-in/zoom-out).
- Horizontal and vertical flipping to simulate different scanning orientations.
- Minor color and contrast adjustments. This process artificially enlarges the effective size of the training pool, which is essential for successfully training very deep architectures.

4.4 Model Architecture Design and Training Strategy

This methodology is based on leveraging the superior feature extraction capabilities of highly effective pre-trained CNN models through **Transfer Learning**.

4.4.1 Selection of Base Models

The project will use two powerful and distinct State-of-the-Art CNN architectures:

- **ResNet50 (Residual Network):** Chosen for its deep structure, which uses 'skip connections' to effectively train many layers without suffering from gradient degradation, thus ensuring comprehensive feature learning.

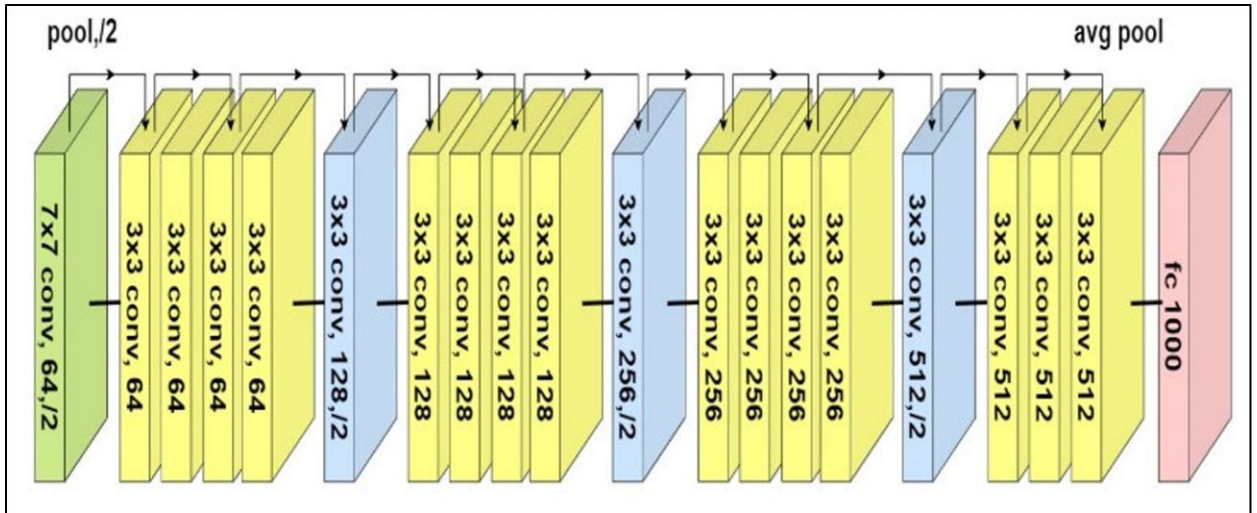


Figure (5): ResNet-50 architecture. [12]

- **EfficientNet-B7:** Selected for its exceptional balance between high accuracy and computational efficiency, achieved through a novel **composite scaling method** that uniformly scales network depth, width, and input resolution.

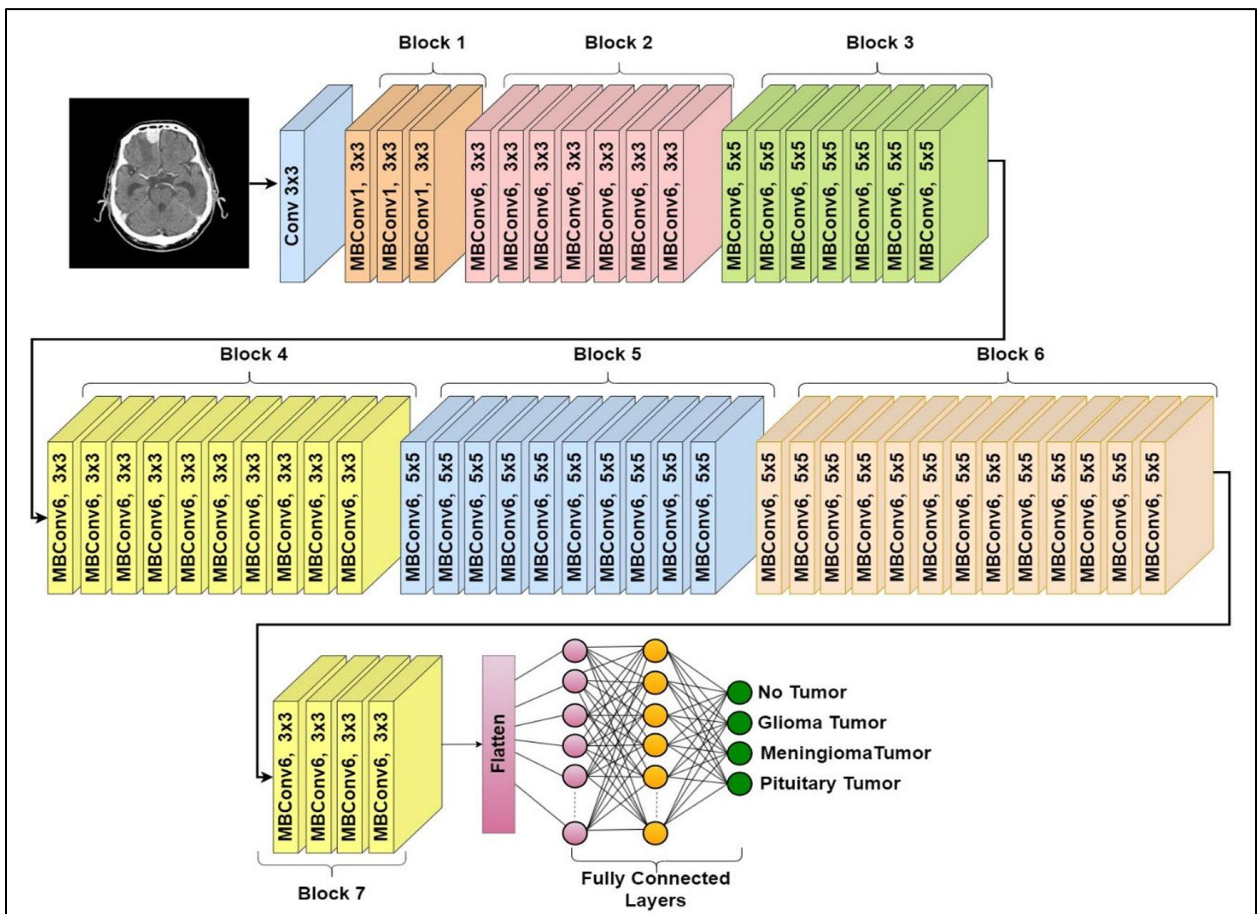


Figure (6): EfficientNet-B7 architecture. [12]

4.4.2 Implementation of Transfer Learning (Fine-Tuning)

Transfer Learning is applied using two high-performance pre-trained CNN architectures: **EfficientNet-B7** and **ResNet-50**. Both models are initialized with ImageNet pre-trained weights to leverage their strong capability in extracting general visual features. Instead of relying on a single architecture, the system adopts an **ensemble approach**, where both models process the MRI images in parallel to obtain deep feature representations. The extracted features from EfficientNet-B7 and ResNet-50 are then combined using a **weighted fusion strategy**, where EfficientNet-B7 contributes **70%** of the weight and ResNet-50 contributes **30%**. This weighted fusion enhances model stability and improves classification performance by balancing the strengths of both architectures. After the fusion stage, the combined feature vector is passed through a sequence of fully connected layers to refine learning for the four target tumor classes. These layers include **Batch Normalization**, a **Dense layer with 256 units and L1/L2 regularization**, followed by a **Dropout layer (0.5)** to reduce overfitting. Finally, a **Softmax output layer** is applied for multi-class prediction. This ensemble-based fine-tuning strategy enables the system to capture richer and more discriminative MRI features, leading to higher accuracy and improved generalization in brain tumor classification.

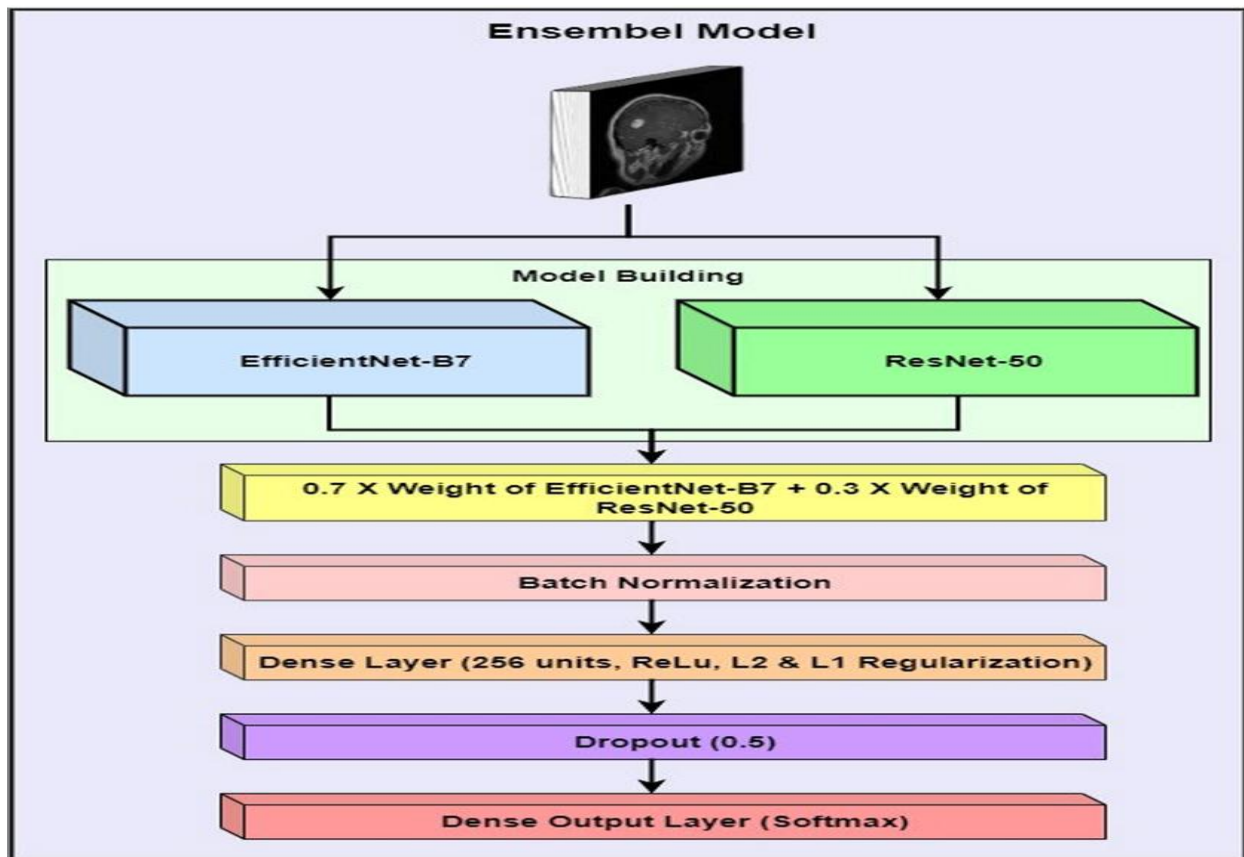


Figure (7): Ensemble model architecture. [12]

4.5 Implementation Details and Performance Metrics

4.5.1 Training Protocol and Environment

The implementation will rely on a specialized Deep Learning platform (such as TensorFlow or PyTorch). Due to the complexity of the models and the high-resolution data, training will be executed on a high-performance computing system equipped with modern GPU units. The training protocol involves selecting an appropriate **Optimizer** (e.g., Adam), defining a suitable **Loss Function** (e.g., Categorical Cross-Entropy), and carefully determining the optimal values for the **Learning Rate** and the number of **Epochs** to ensure efficient convergence.

4.5.2 Evaluation Metrics

The final evaluation of the system's performance will be quantified using a set of well-established metrics. The primary measure is **Classification Accuracy**. Secondary metrics will include **Precision, Recall, and the F1-Score**. These additional metrics are crucial for gaining a complete understanding of the model's diagnostic strengths, particularly in assessing its ability to correctly identify positive cases (tumors) while minimizing diagnostic errors.

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